HDOC DAY – 60

Algorithm

1. Start.
2. Read strings s and t.
3. Pass their addresses to check_anagram().
4. Compare the lengths of both strings.
5. If the lengths are different, display "Not Anagram" and stop the function.
6. Initialize a frequency array of size 26 with all values as 0.
7. Traverse both strings.
8. Increase the frequency for each character of s.
9. Decrease the frequency for each character of t.
10. Traverse the frequency array.
11. If any frequency is not 0, display "Not Anagram".
12. Otherwise, display "Anagram".
13. Stop.

C PROGRAM:

```
#include <stdio.h>
#include <string.h>

void check_anagram(char *s, char *t)
{
   int freq[26] = {0};
   int i;
```

```
    if (strlen(s) != strlen(t))
    {
        printf("Not Anagram\n");
        return;
    }

    for (i = 0; *(s + i) != '\0'; i++)
    {
        freq[*(s + i) - 'a']++;
        freq[*(t + i) - 'a']--;
    }

    for (i = 0; i < 26; i++)
    {
        if (freq[i] != 0)
        {
            printf("Not Anagram\n");
            return;
        }
    }

    printf("Anagram\n");
}

int main()
```

```
{
    char s[100], t[100];

    printf("Enter string s: ");
    scanf("%s", s);

    printf("Enter string t: ");
    scanf("%s", t);

    check_anagram(s, t);

    return 0;
}
```

TEST CASE TABLE:

Test Case	String s	String t	Expected Output
1	listen	silent	Anagram
2	hello	world	Not Anagram
3	anagram	nagaram	Anagram
4	rat	car	Not Anagram
5	abc	ab	Not Anagram
6	aabbcc	abcabc	Anagram